

Scout S. Heid

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WORK EXPERIENCE

Halo Industries, Inc.

2023 – Current

Staff Mechanical Engineer

- Lead Designer for 75% of all motion systems in a new generation laser tool, utilizing pneumatics, valves, solenoids, and sensors.
- Lead Engineer for an 8 motor mirror mount light bridge that reduced calibration time of our high powered lasers by **90%**
- Built a high precision testing fixture for the main optics module within 8 weeks and used to assemble and calibrate 16-beam laser.
- Integrated safety systems, including Keyence interlocks and light curtain monitoring, ensuring compliance with **OSHA** machine safeguarding guidelines, **NFPA79** electrical standards for industrial machines, and **EN954-1/ISO 13849-1** safety and performance.
- Qualified micron step lens motor's accuracy, precision and bi-directional error with my own custom test fixture that improved kerf by 20%. **NewScale API** with **python** and integrated with Keyence **CL-3000** controller and **CL-P015N** confocal sensor plus optical unit.

Amazon Lab126, Advanced Products

2019 – 2023

Product Design Engineer

- Lead product design engineer for dynamic motion sub-assy on future products. DRI for 15+ parts and 5+ sub-assembly. Reduced cost by **30%**; Part count by **25%**, and noise dB by **50%** with a modularized tilt mechanism for flexibility along the entire design process.
- DRI for 2 noise reduction parts in the Astro pan sub-system. Reduced dB by 20%, resulting in passed audio requirement.
- Lead PD for NTI of low cost/noise/backlash, high torque/stiffness gearmotors for robotic actuators with novel integrated torque/force sensing. Reduced cost by 80% of market samples and characterized KPI range for use in the home.
- Evaluated user studies internally and by Stanford for home robotics use cases and prototyped initial products for market.
- Developed a simultaneously sequenced mast for high load lifting using innovative leadscrew gear train system to simplify complex Astro lift mechanism. Greatly reduced the complexity and removed current device primary failure modes.

Product Design Engineer, R&D Intern

May 2018 – Aug 2018

- Prototyped concept development robot and ran a functional demo of 6 DOF dynamic devices for VP & Director.
- Invented 2 proprietary gripper mechanisms specialized for home environment assistance and synced with the 6 DOF system.
- Utilized ROS, wifi, serial, and firmware code to connect a network of complex systems and sensors that control and vision.

Apple Inc.

Jan 2017 – Aug 2017

iPhone Power PD, Intern

- Lead PD engineer for architecture and development of JDM program with 3rd party vendor.
 - Reviewed DFMs, mold flow analysis, and overall design of the product. Enforced strict safety and design guidelines.
 - Mixture of design verification/testing, and managing communicating with China team to implement improvements.
- Oversaw the entire assembly, product line, and processes for parts I owned through the design/production cycle, including China fab.
- Researched optimized ultrasonic welding design using various cosmetic and weld quality metrics. Tested with DOE
- Communicated daily with cross functional teams (safety, EE, vendors) to remove blockers for all JDM components.

Curtis Instruments

May 2016 – Aug 2016

Mechanical Engineer, Intern

- Constructed a software architecture that added 10+ separately controlled devices to a dynamic User Interface in LABVIEW.
 - Used asynchronous calls, user events, and references to control devices in a unique variation of Actor Framework.
 - Wrapped existing code and reprogrammed devices for optimal efficiency and speed when attaining and sending data.
- Operated and tested on multiple testing devices, including a Dynamometer, Power Supply, DMM, Oscilloscope and the company's Controller, which used a variety of interfacing methods (GPIB, USB, Serial, CANopen, VISA).

LEADERSHIP

Halo Industries - Head engineer of the intern program and managing 2-3 interns throughout the year with 70% return rate.

Amazon - Manager & mentor for Summer 2022 interns across the org (PD/EE/Controls). Led weekly presentations and team events.

EDUCATION

University of California, Berkeley

Graduated 2019

Master's of Engineering, Controls

Focus: Autonomous Vehicle Controls + Path Planning

B.S Mechanical Engineering / Minor in EECS

Focus: Product Design + Robotics/Manipulation

ACCOLADES

UC Berkeley Pioneers in Engineering (PiE) – UC Berkeley <i>Technical Director (PiE)</i>	Mar 2015 – May 2016
Lab Assistant, Lab Developer & Reader – UC Berkeley	Jan 2015 – Jan 2016
Pi Tau Sigma Engineering Honor Society (PTS)	Aug 2014 – Dec 2018
Lab Researcher for Biomimetic Millisystems Lab	Feb 2014 – Dec 2016

SKILLS

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- Programming Languages: Java, Python, ROS, Arduino, MATLAB, Simulink, LabView, HTML & CSS, Ruby, LaTeX C++
 - CAD/CAM: SolidWorks, Autodesk Inventor, AutoCAD, Fusion 360, Mastercam, EAGLE, COMSOL, NX10, Creo, KISSys
 - Skills: Injection molding, Die-cut, Diecasting, GD&T, Tolerance analysis, Machine Shop, Circuit Board Production, Reflow Soldering, Casting, Milling, Laser Cutter, Microsoft Office, Git, Design of experiments, Design for manufacturability, Design for Assembly, Failure Analysis, Robotics, Baxter, Gearbox design, High volume and mass production manufacturing, pneumatics, safety interlocks, statistical process control (SPC), Keyence.
 - Languages: English and fluent in conversational Japanese.